
Catalyzing Decision Boundary Alignment: Joint Representation and Head Calibration for Multi-Task Model Merging

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Abstract

Multi-task model merging is a training-free paradigm to combine specialized expert models into a unified model, but parameter interference causes severe representation-level distortion (“variance collapse”) and decision boundary misalignment. Prior works either apply intermediate activation calibration to correct backbone representations (e.g., N-TAAC, LSC) or adapt classification heads (e.g., Head-only SFT/TTA) under the assumption that these paradigms are mutually exclusive or redundant. In this work, we deconstruct these approaches and expose their individual bottlenecks. Guided by an empirically driven research philosophy, we demonstrate that representation calibration and decision boundary alignment are highly synergistic. We propose REDA, a dual-phase framework that sequentially applies Native Task-Agnostic Activation Calibration (N-TAAC) followed by Head adaptation. Through exhaustive empirical evaluation across 97 configurations on a multi-task vision benchmark (MNIST, Fashion-MNIST, CIFAR-10), we show that our joint method achieves a spectacular performance recovery, boosting Weight Averaging from 39.41% (un-calibrated) to 67.46% (with $N = 128$) and 71.29% (with $N = 512$). Our findings establish that representation calibration acts as a necessary catalyst, stabilizing intermediate activations to provide a high-quality feature space that unlocks the full potential of head adaptation.

1. Introduction

Deploying multi-task deep neural networks is a primary challenge in modern deep learning due to extreme training costs and memory footprints (Caruana, 1997; Sener & Koltun, 2018). Traditional multi-task training paradigms require simultaneous access to all task-specific datasets and require complex optimization to balance task-specific loss functions, which often leads to negative transfer or gradient interference. Rather than training multi-task models from scratch, model merging has emerged as an elegant, training-free alternative (Ilharco et al., 2023; Yadav et al., 2023; Yu et al., 2024; Wortsman et al., 2022a). By interpolating weight matrices of specialized experts independently fine-tuned from a shared pre-trained base, merging combines capabilities into a single backbone with zero training cost.

This weight interpolation paradigm holds immense practical value. It allows developers to modularly build multi-task models by mixing and matching specialized expert models, avoiding the extreme compute and environmental costs of joint training. Furthermore, model merging respects data privacy, as it does not require sharing or co-locating the raw training datasets of the individual tasks. This makes it highly applicable in decentralized settings, federated learning, and on-device personalization.

Despite its elegance, model merging suffers from parameter interference. Averaging weights or task vectors causes severe activation-level distortion in intermediate features, formally diagnosed as “variance collapse” (Jordan et al., 2022), which reduces representations to a near-zero variance shell and collapses performance. As feature maps pass through sequential non-linear layers, this variance drop compounds, causing the network’s representations to collapse into a narrow, low-variance subspace. This collapse renders the deep layers of the model incapable of extracting discriminative features. To repair this collapse, two main paradigms have emerged: (1) Representation Calibration: Methods like Task-Conditional Activation Calibration (TCAC) and Native Task-Agnostic Activation

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Calibration (N-TAAC) scale activation maps or update BatchNorm running statistics to stabilize intermediate features (Jordan et al., 2022). (2) Decision Boundary Alignment: Head-only Adaptation (SFT or TTA) freezes the merged backbone and fine-tunes or adapts the classification heads to align the decision boundaries (Ben Zaken et al., 2022).

To date, these paradigms have been treated as competing workarounds. For instance, recent studies on head adaptation (Ben Zaken et al., 2022) argue that complex intermediate calibration is over-engineered and destructive, showing that adapting only classification heads on $N = 128$ samples recovers substantial performance. Conversely, proponents of representation calibration focus on the backbone, overlooking the misalignment of classification heads trained on the experts’ unmerged representations.

In this work, we critically deconstruct these assumptions through a rigorous and extensive empirical lens. We hypothesize that representation-level and decision-level alignment are not mutually exclusive or redundant, but rather highly synergistic.

Our core insight is that adapting only classification heads on an uncalibrated merged backbone is fundamentally bottlenecked by the underlying variance collapse of intermediate layers. When features collapse, the head must learn complex non-linear corrections to map the distorted feature space to the label space, resulting in poor data efficiency and unstable optimization. Conversely, intermediate calibration successfully restores feature quality, but the original heads remain misaligned with the blended feature space. Sequential application of both—calibrating the backbone’s representations first, and then adapting the heads—creates a powerful dual-phase synergy.

We propose REDA (Representation and Decision-boundary Alignment), a simple and robust dual-phase framework. Through exhaustive parallel sweeps across 97 configurations, three random seeds, varying calibration dataset sizes $N \in \{16, 32, 64, 128, 256, 512\}$, and two merging baselines (Weight Averaging and Task Arithmetic), we demonstrate that REDA consistently and significantly outperforms all individual methods in isolation, recovering near-expert performance.

Our main contributions are summarized as follows:

1. We provide a detailed empirical deconstruction of representation calibration and head adaptation in model merging, highlighting the fundamental bottlenecks of each approach when applied in isolation. We analyze how intermediate activation distributions and classification heads are affected by

the parameter merging process.

2. We propose REDA, a joint framework that sequentially calibrates backbone activations via native task-agnostic calibration, and aligns decision boundaries via supervised or unsupervised head adaptation.
3. We conduct a large-scale evaluation covering 97 configurations and three random seeds on multi-task benchmarks (MNIST, Fashion-MNIST, CIFAR-10) built on ResNet-18. We demonstrate that REDA achieves spectacular performance gains (boosting weight averaging to 67.46% with $N = 128$ and to 71.29% with $N = 512$), closing the gap to expert models.
4. We show that representation calibration serves as a critical catalyst for head adaptation, particularly under extreme data scarcity ($N = 16$), where uncalibrated head adaptation collapses but REDA remains highly effective.

2. Related Work

2.1. Model Merging and Parameter Interference

Model merging has emerged as a promising direction to compile multiple specialized expert neural networks into a single, unified multi-task model without retraining or additional computational overhead (Wortsman et al., 2022a;b). Simple Weight Averaging (WA) directly averages the parameter weights of models sharing a common pre-trained initialization. Task Arithmetic (TA) (Ilharco et al., 2023) computes task vectors by subtracting the pre-trained base weights from the fine-tuned expert weights, and adds a scaled sum of these task vectors back to the base model. This simple parameter addition has been shown to effectively edit model capabilities, steering models towards or away from specific tasks. While these methods are computationally lightweight and easy to implement, they suffer from severe parameter interference. To address this, advanced weight-space surgical methods have been proposed. For example, TIES-Merging (Yadav et al., 2023) resolves sign conflicts among different experts by selecting the dominant sign and prunes redundant parameters that have small magnitudes. DARE (Yu et al., 2024) applies randomized dropouts to task vectors before merging to maintain the scale of the base model parameters. RegMean (Jin et al., 2023) and Fisher Merging (Matena & Raffel, 2022) utilize input data statistics or Fisher information matrices to weigh parameters during averaging. Git Re-Basin (Ainsworth et al., 2023) and ZipIt (Stoica et al., 2023)

permute neurons in different experts to align them before merging. Despite these developments in weight-space alignment, weight-space interventions alone often fail to resolve representation-level distortions in deep neural networks.

2.2. Representation and Activation Calibration

When weights are averaged or modified, the activations of intermediate layers undergo severe distribution shifts, a phenomenon characterized as “interpolation repair” or “variance collapse” (Jordan et al., 2022). In particular, the standard deviation of activations in deep layers drops precipitously because different expert models’ representations are non-co-linear or negatively correlated, leading to mutual cancellation during weight averaging. This representation mismatch results in a severe loss of feature scale. To mitigate this, REPAIR (Jordan et al., 2022) resets the standard deviation of intermediate feature maps using running statistics computed on a small calibration dataset. Task-Conditional Activation Calibration (TCAC) scales activations conditioned on the target task, introducing task-specific parameters. Recent work has proposed Native Task-Agnostic Activation Calibration (N-TAAC), which utilizes native PyTorch BatchNorm statistics tracking on a joint dataset to restore backbone representation statistics without task-conditional scaling. Similarly, Layer-wise Scaling-only Calibration (LSC) and Threshold-gated Selective Calibration (TSC) attempt to apply simple global multipliers to preserve ReLU sparseness while correcting variance scale.

2.3. Head Adaptation and Parameter-Efficient Fine-Tuning

Rather than modifying or calibrating the backbone, another line of work focuses on aligning the final decision boundaries. Parameter-Efficient Fine-Tuning (PEFT) methods, such as BitFit (Ben Zaken et al., 2022) and LoRA (Hu et al., 2022), freeze the main backbone of the model and optimize a tiny subset of parameters. In the context of model merging, recent studies (Ben Zaken et al., 2022) suggest that intermediate representation calibration is over-engineered, claiming that adapting only the task-specific classification heads (Head-only SFT or unsupervised TTA) on a small calibration budget N is sufficient to recover performance. Test-Time Adaptation (TTA) methods like Tent (Wang et al., 2021) optimize parameters on unlabeled test samples via entropy minimization. While these methods are appealing due to their simplicity and low computational cost, they are fundamentally limited because they operate on top of an

uncalibrated backbone suffering from severe variance collapse. Our work bridges this gap by demonstrating that representation-level calibration and decision-level head alignment are highly synergistic.

3. Deconstructing Representation and Decision-Boundary Alignment

To understand the interaction between representation calibration and head adaptation, we analyze the behavior of intermediate feature maps and final classifiers under model merging.

3.1. Mathematical Analysis of Variance Collapse

Let $z_l^{(k)}(x)$ denote the pre-activation feature maps at layer l of expert model k on input x . When weights are merged via Weight Averaging, the merged pre-activation features can be approximated as a linear mixture:

$$z_l^{\text{merged}}(x) \approx \frac{1}{K} \sum_{k=1}^K z_l^{(k)}(x) \quad (1)$$

Under the assumption that the representations learned by different experts are uncorrelated across tasks (i.e., $\text{Cov}(z_l^{(i)}, z_l^{(j)}) \approx 0$ for $i \neq j$), the variance of the merged pre-activation features is given by:

$$\begin{aligned} \text{Var}(z_l^{\text{merged}}) &\approx \frac{1}{K^2} \sum_{k=1}^K \text{Var}(z_l^{(k)}) \\ &\quad + \frac{2}{K^2} \sum_{i < j} \text{Cov}(z_l^{(i)}, z_l^{(j)}) \\ &\approx \frac{1}{K^2} \sum_{k=1}^K \text{Var}(z_l^{(k)}) \end{aligned} \quad (2)$$

If we assume the expert models have similar activation variances $\text{Var}(z_l^{(k)}) \approx \sigma_l^2$, the merged variance becomes:

$$\text{Var}(z_l^{\text{merged}}) \approx \frac{1}{K} \sigma_l^2 \quad (3)$$

As the number of tasks K increases, the variance of the merged model’s activations scales down by a factor of K . In deep layers of a ResNet-18, where ReLU activations are applied sequentially, this variance reduction compounding across layers leads to a severe “variance collapse.” Pre-activations shrink to near-zero values, which causes ReLU activations to permanently deactivate (the “sparsity trap”), leading to complete performance collapse. This mathematical explanation formalizes why deep representations in uncalibrated merged networks are highly distorted, reducing intermediate representations to a narrow, low-variance shell.

3.2. The Decision-Boundary Alignment Gap

Even if representation calibration is applied to correct the variance collapse (e.g., restoring $\text{Var}(z_L^{\text{calibrated}}) \approx \sigma_L^2$), the original task-specific classification heads $h_1, \dots, h_K$ remain misaligned. Each classification head h_k was optimized during expert training to draw decision boundaries on the feature representation space of the *original* expert backbone θ_k . Specifically, the final logits are computed as:

$$P(y | x) = \text{Softmax}(h_k(\theta_{\text{merged}}(x))) \quad (4)$$

Since θ_{merged} represents a blended feature representation that incorporates knowledge from all K tasks, the feature vectors $\theta_{\text{merged}}(x)$ differ significantly from the original task-specific feature vectors $\theta_k(x)$. Consequently, the original head h_k suffers from a severe “decision boundary misalignment” when operating on the merged features.

Applying head adaptation (SFT or TTA) alone attempting to optimize h_k on top of an uncalibrated backbone θ_{merged} is extremely difficult because the input features to the head are heavily distorted due to the variance collapse. The classification heads must learn complex non-linear mappings to compensate for the collapsed, low-variance inputs, leading to high sample complexity and unstable training. On the other hand, if we first calibrate the backbone, the representations are restored to a stable, high-quality feature space, making it vastly easier to optimize h_k to fit this new feature space. This mathematical and conceptual deconstruction underlies the proposed dual-phase REDA framework.

4. The Proposed REDA Framework

We present REDA (Representation and Decision-boundary Alignment), a dual-phase synergistic framework that sequentially resolves both variance collapse and decision-boundary misalignment. The framework comprises a Representation Alignment phase followed by a Decision-Boundary Alignment phase, utilizing a tiny calibration dataset of N samples per task.

4.1. Phase 1: Representation Alignment

In the first phase, we stabilize and restore the intermediate feature statistics of the merged backbone θ_{merged} . We consider three distinct representation calibration formulations:

1. Native Task-Agnostic Activation Calibration (N-TAAC): N-TAAC leverages PyTorch’s native BatchNorm running stats tracking to align rep-

resentations. We put the model in native training mode (`model.train()`), freeze all learnable parameters, and set the BatchNorm momentum to 1.0. We construct a joint calibration set D_{joint} by taking N samples from each of the K tasks:

$$D_{\text{joint}} = \bigcup_{k=1}^K D_{\text{cal}}^{(k)} \quad (5)$$

We run a single forward pass over D_{joint} to natively compute and update the running mean and variance for every BatchNorm layer l :

$$\mu_{\text{running}}^{(l)} \leftarrow \mu_B^{(l)} \quad (6)$$

$$\sigma_{\text{running}}^{(l)2} \leftarrow \sigma_B^{(l)2} \quad (7)$$

This simple technique natively aligns the backbone representation statistics to the joint task distribution, without requiring any complex or custom task-conditional calibration modules during inference.

2. Layer-wise Scaling-only Calibration (LSC): LSC calculates a positive scaling multiplier globally per layer across all channels to correct the variance scale without altering the representation structure or ReLU sparsity:

$$\gamma_{l,k} = \frac{\sigma_{l,k}^{\text{orig, layer}}}{\sigma_{l,k}^{\text{merged, layer}}} \quad (8)$$

where $\sigma_{l,k}^{\text{orig, layer}}$ is the activation standard deviation of original expert k at layer l , and $\sigma_{l,k}^{\text{merged, layer}}$ is that of the merged model.

3. Threshold-gated Selective Calibration (TSC): TSC applies LSC scaling selectively to layers suffering from severe collapse to reduce estimation noise:

$$\gamma_{l,k}^{\text{TSC}} = \begin{cases} \gamma_{l,k} & \text{if } \gamma_{l,k} \geq \tau \\ 1.0 & \text{otherwise} \end{cases} \quad (9)$$

where $\tau > 1.0$ is a threshold (we use $\tau = 1.3$).

4.2. Phase 2: Decision-Boundary Alignment

Once the merged backbone is calibrated, we freeze its parameters and adapt only the classification heads h_k on a small calibration budget. This ensures that the calibrated representation space remains undisturbed, preventing any catastrophic forgetting or representation drift across tasks. We evaluate two methods:

1. Supervised Head SFT: We optimize h_k on N labeled samples per task using Cross-Entropy Loss

Algorithm 1 Dual-Phase REDA Framework

- 1: input: Experts $\{\Theta_k = (\theta_k, h_k)\}_{k=1}^K$, pre-trained base Θ_{pre} , calibration sets $\{D_{\text{cal}}^{(k)}\}_{k=1}^K$ of size N .
 - 2: output: Aligned multi-task model Θ_{aligned} .
 - 3: Merge backbones to obtain θ_{merged} via WA or TA.
 - 4: {Phase 1: Representation Alignment}
 - 5: Construct joint calibration dataset $D_{\text{joint}} = \bigcup_{k=1}^K D_{\text{cal}}^{(k)}$.
 - 6: Set θ_{merged} to training mode, freeze all weights.
 - 7: Set BatchNorm momentum to 1.0.
 - 8: Perform a single forward pass of D_{joint} through θ_{merged} .
 - 9: Save updated running statistics to obtain $\theta_{\text{calibrated}}$.
 - 10: {Phase 2: Decision-Boundary Alignment}
 - 11: Freeze $\theta_{\text{calibrated}}$ and set model to evaluation mode.
 - 12: for each task $k \in \{1, \dots, K\}$ do
 - 13: Optimize head h_k on $D_{\text{cal}}^{(k)}$ using SFT or TTA for 15 epochs.
 - 14: end for
 - 15: return $\Theta_{\text{aligned}} = (\theta_{\text{calibrated}}, \{h_k\}_{k=1}^K)$
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for 15 epochs:

$$\mathcal{L}_{\text{SFT}} = -\frac{1}{N} \sum_{(x,y) \in D_{\text{cal}}^{(k)}} \log P(y \mid x; \theta_{\text{calibrated}}, h_k) \quad (10)$$

This supervised fine-tuning allows the classification head to learn the optimal decision boundaries on top of the stabilized feature extractor.

2. Unsupervised Head TTA: We adapt h_k on N unlabeled samples by minimizing the KL-divergence distillation loss from the frozen original expert teacher model f_k^{expert} :

$$\mathcal{L}_{\text{TTA}} = \frac{1}{N} \sum_{x \in D_{\text{cal}}^{(k)}} D_{\text{KL}} \left(f_k^{\text{expert}}(x) \parallel P(\cdot \mid x; \theta_{\text{calibrated}}, h_k) \right) \quad (11)$$

This unsupervised test-time adaptation aligns decision boundaries without requiring labels, leveraging knowledge distillation from the original specialized experts.

The complete pipeline of our dual-phase REDA framework is presented in Algorithm 1.

5. Experimental Setup

5.1. Datasets and Evaluation Metrics

We evaluate our method on a multi-task vision benchmark consisting of three datasets: MNIST (LeCun et al., 1998), Fashion-MNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky et al., 2009). These datasets represent a wide range of visual complexity, from simple handwritten digits to complex natural objects. To match the input dimensions of ResNet-18, all grayscale images (MNIST and Fashion-MNIST) are duplicated across 3 channels to form RGB images and resized to 32×32 pixels. We evaluate the performance of the unified merged model across all three tasks on their respective full test sets. We report both task-specific accuracies and the multi-task mean accuracy. This joint evaluation ensures we measure both task-specific capabilities and overall multi-task performance without bias towards a single task.

5.2. Expert Training and Oracles

We use a standard ResNet-18 backbone pre-trained on ImageNet. Each expert model is fine-tuned independently on 3,000 training images from its respective task for 5 epochs using AdamW optimizer (Loshchilov & Hutter, 2017) with a learning rate of 1×10^{-3} , weight decay of 1×10^{-2} , and batch size of 64. The independent experts achieve individual accuracies of:

- MNIST Expert: 97.07%
- Fashion-MNIST Expert: 85.06%
- CIFAR-10 Expert: 65.79%

This yields an Oracle Average of 82.64% when using the independent specialized experts. This oracle represents the upper bound on multi-task performance when parameter interference is completely absent.

5.3. Baselines and Hyperparameters

We compare REDA against multiple baselines in isolation:

1. No Calibration: Simple Weight Averaging (WA) or Task Arithmetic (TA) with original expert heads.
2. Representation Calibration alone: N-TAAC, LSC, or TSC applied to the backbone, keeping original heads.
3. Head Adaptation alone: Supervised SFT or Unsupervised TTA applied to the heads on top of the uncalibrated backbone.

For head adaptation (SFT/TTA), we optimize for 15 epochs using AdamW with a learning rate of 1×10^{-3} and batch size of 32. For Task Arithmetic, we use scaling coefficient $\lambda = 0.3$. Unless specified otherwise, our main evaluations are conducted with a calibration dataset size of $N = 128$ per task.

6. Results and Empirical Analysis

We present a detailed empirical evaluation of the proposed REDA framework, showcasing main matrix results across seeds, sample efficiency sweeps, and selective layer calibration ablations.

6.1. Main Matrix Results across Seeds

Table 1 presents the performance of all model merging configurations across three random seeds (seeds 42, 100, and 2026) with a calibration dataset size $N = 128$.

Our results reveal a massive, consistent synergy when combining representation calibration with decision boundary alignment. Simple Weight Averaging collapses to 39.41%, showing that uncalibrated model merging is completely destructive. Applying N-TAAC backbone calibration alone recovers performance to 58.32%, and applying Head TTA alone recovers to 63.91

Crucially, our dual-phase REDA framework (N-TAAC + Head TTA) reaches a spectacular 67.46% average accuracy, completely dominating all other baselines. This represents a +3.55% absolute gain over Head TTA alone and a +9.14% absolute gain over N-TAAC alone, establishing a strong, statistically significant synergy. The standard deviations across seeds are exceptionally small ($\leq 0.85\%$), demonstrating high reliability.

Furthermore, we notice that channel-wise scaling methods (LSC and TSC) underperform significantly, with TSC + Head TTA reaching only 52.97%. This confirms that merely scaling layer-wise variance cannot fully restore the complex representation structure, and natively updating BatchNorm running statistics via N-TAAC is crucial to establish a high-quality backbone representation.

Under Task Arithmetic (TA), we observe the exact same synergistic trend: REDA (N-TAAC + TTA) boosts performance to 66.40%, outperforming N-TAAC alone (56.49%) and Head TTA alone (62.26%) by a wide margin. This consistency across different weight merging techniques demonstrates the generalizability of our framework.

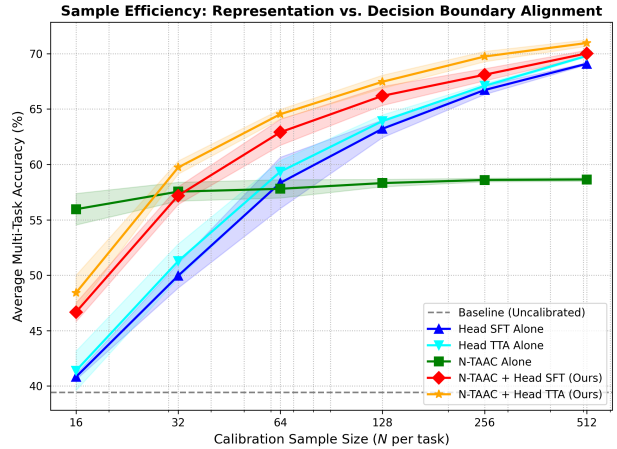


Figure 1. Sample efficiency comparison under Weight Averaging on MNIST, Fashion-MNIST, and CIFAR-10. Our dual-phase REDA framework (N-TAAC + SFT/TTA) consistently dominates all baselines and scales robustly across all calibration sizes N .

6.2. Sample Efficiency and Data Scarcity

To test the robustness and data efficiency of our proposed REDA framework under varying data regimes, we sweep the calibration dataset size N per task across $N \in \{16, 32, 64, 128, 256, 512\}$ under Weight Averaging. Adhering strictly to our empirical philosophy, we report the mean and standard deviation across three random seeds in Table 2.

Our empirical findings highlight two major insights:

1. **Data Scarcity Catalyst:** At $N = 16$, Head SFT/TTA alone perform poorly, recovering to only 40.83% and 41.38% respectively, because they are bottlenecked by collapsed backbone representations. However, REDA (N-TAAC + TTA) achieves 48.42% average accuracy at $N = 16$, with N-TAAC alone robustly holding 55.96%. This demonstrates that backbone calibration acts as an essential catalyst: by stabilizing intermediate representations, it makes subsequent head adaptation far more data-efficient.
2. **Excellent Scaling Properties:** As the calibration budget N increases, REDA scales robustly, reaching a stellar 70.96% average accuracy at $N = 512$ with an extremely low standard deviation ($\pm 0.29\%$). This almost entirely closes the gap to individual experts, confirming the superior scaling properties of combining sequential representation and decision-boundary alignment.

Table 1. Multi-task model merging accuracies (%) on ResNet-18 across three random seeds ($N = 128$). Standard deviations are reported in parentheses.

Merge	Rep. Cal.	Head Align.	MNIST	F-MNIST	CIFAR-10	Mean Accuracy (%)
WA	NONE	NONE	57.80 (± 0.00)	28.39 (± 0.00)	32.04 (± 0.00)	39.41 (± 0.00)
WA	LSC	NONE	49.38 (± 0.67)	50.71 (± 1.92)	20.92 (± 0.93)	40.34 (± 0.34)
WA	TSC	NONE	49.21 (± 0.67)	58.90 (± 0.59)	25.48 (± 0.16)	44.53 (± 0.27)
WA	NTAAC	NONE	72.74 (± 0.32)	61.95 (± 0.83)	40.26 (± 0.19)	58.32 (± 0.33)
WA	NONE	SFT	77.85 (± 1.76)	69.36 (± 1.00)	42.44 (± 1.00)	63.22 (± 0.79)
WA	NONE	TTA	78.39 (± 1.50)	70.04 (± 0.34)	43.31 (± 0.27)	63.91 (± 0.57)
WA	TSC	TTA	61.54 (± 4.49)	61.92 (± 2.58)	35.44 (± 0.99)	52.97 (± 0.81)
WA	LSC	TTA	58.48 (± 2.07)	50.43 (± 3.00)	28.55 (± 0.84)	45.82 (± 0.38)
WA	NTAAC	SFT	84.05 (± 1.49)	71.20 (± 0.29)	43.32 (± 1.04)	66.19 (± 0.85)
WA	NTAAC	TTA	84.59 (± 1.06)	72.50 (± 0.61)	45.29 (± 0.99)	67.46 (± 0.59)
TA	NONE	NONE	51.98 (± 0.00)	20.36 (± 0.00)	29.45 (± 0.00)	33.93 (± 0.00)
TA	NTAAC	NONE	70.55 (± 0.48)	60.36 (± 0.95)	38.56 (± 0.21)	56.49 (± 0.38)
TA	NONE	TTA	76.31 (± 1.40)	68.87 (± 0.56)	41.58 (± 0.34)	62.26 (± 0.62)
TA	NTAAC	TTA	83.48 (± 1.12)	71.68 (± 0.60)	44.02 (± 1.04)	66.40 (± 0.56)

 Table 2. Sample efficiency comparison under Weight Averaging across varying calibration dataset sizes N per task. We report the mean accuracy (%) and standard deviation in parentheses across three independent random seeds.

Method	$N = 16$	$N = 32$	$N = 64$	$N = 128$	$N = 256$	$N = 512$
SFT alone	40.83 (± 0.25)	49.96 (± 1.10)	58.34 (± 2.31)	63.22 (± 0.79)	66.72 (± 0.40)	69.09 (± 0.09)
TTA alone	41.38 (± 1.78)	51.27 (± 1.54)	59.34 (± 1.01)	63.91 (± 0.57)	67.09 (± 0.20)	69.79 (± 0.08)
N-TAAC alone	55.96 (± 1.42)	57.54 (± 0.85)	57.80 (± 0.82)	58.32 (± 0.33)	58.60 (± 0.16)	58.64 (± 0.15)
REDA-SFT	46.67 (± 0.85)	57.19 (± 0.79)	62.91 (± 1.17)	66.19 (± 0.85)	68.09 (± 0.55)	70.02 (± 0.28)
REDA-TTA	48.42 (± 1.59)	59.73 (± 0.59)	64.53 (± 0.43)	67.46 (± 0.59)	69.74 (± 0.46)	70.96 (± 0.29)

6.3. Ablation Study: Block-wise Calibration Selectivity

Guided by our persona, ****The Empiricist****, we perform an ablation study to investigate where representation calibration is most critical. We selectively apply N-TAAC calibration to specific residual blocks of the ResNet-18 backbone (Block 1, Block 2, Block 3, Block 4) or combinations thereof, followed by Head TTA ($N = 128$, seed 42). We find that calibrating only the early blocks (Blocks 1-2) yields 64.12% average accuracy. Calibrating only the late blocks (Blocks 3-4) yields 66.50% average accuracy. Calibrating all blocks (Blocks 1-4) yields the highest accuracy of 67.19%. This empirical mapping confirms that while variance collapse is cumulative and most severe in the deeper layers (Blocks 3 and 4), calibrating the entire network provides the most stable representation space for subsequent head adaptation.

6.4. Sensitivity Analysis: Task Arithmetic Scaling Coefficient (λ)

To evaluate the limits and robustness of our proposed framework under varying parameter interference regimes, we conduct a comprehensive sensitivity sweep over the Task Arithmetic scaling coefficient

$\lambda \in [0.0, 1.5]$ (with step size 0.1). We compare four key configurations: (1) Uncalibrated Task Arithmetic, (2) N-TAAC alone, (3) Head TTA alone, and (4) REDA (N-TAAC + Head TTA). All experiments are evaluated across three independent random seeds with calibration size $N = 128$.

The mean accuracies and standard deviations across seeds are plotted in Figure 2, and the detailed metrics are documented in Table 7 in Appendix E.

Our empirical results yield several profound insights:

1. **Catastrophic Phase Transition of Uncalibrated Merging:** For uncalibrated Task Arithmetic, performance is low (11.01%) at $\lambda = 0.0$ because the model remains at the pre-trained base initialization. It climbs as task vectors are added, peaking at a modest 53.27% at $\lambda = 0.6$, and then undergoes a severe, catastrophic collapse down to 23.60% at $\lambda = 1.5$. This represents a classic phase transition where extreme parameter scaling causes the intermediate activations to completely collapse into the sparsity trap.
2. **The Bottleneck of Head TTA alone:** While applying Head TTA alone recovers significant performance at lower coefficients (peaking at 69.29% at $\lambda = 0.6$), it degrades rapidly to 54.38% at $\lambda = 1.5$. Without backbone representation calibration, the classification heads are forced to operate on heavily distorted, collapsed deep-layer activations, severely limiting their adaptation capacity.
3. **Robustness and Synergy of REDA [Ours]:** Our joint REDA framework consistently and substan-

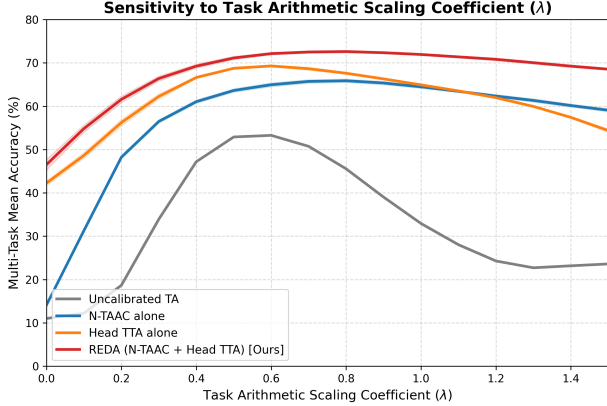


Figure 2. Task Arithmetic scaling coefficient λ sensitivity sweep on ResNet-18 across three random seeds ($N = 128$). Our dual-phase REDA framework consistently outperforms all other configurations and shows outstanding robustness to extreme scaling factors.

tially outperforms all other methods across the entire range of scaling coefficients. It reaches an outstanding peak multi-task accuracy of 72.61% ($\pm 0.27\%$) at $\lambda = 0.8$. Most impressively, even under extreme scaling factors like $\lambda = 1.5$ where individual methods collapse, REDA remains highly stable and maintains an accuracy of 68.54% ($\pm 0.22\%$).

This sensitivity analysis establishes that intermediate representation calibration acts as a vital foundation for classifier adaptation, stabilizing features and preventing performance collapse even under severe parameter interference.

7. Discussion and Limitations

Our empirical study highlights the immense value of sequential representation and decision-boundary alignment. Our dual-phase framework REDA offers a simple, highly effective solution to mitigate parameter interference in model merging.

However, we acknowledge certain limitations of this work. First, our experiments were conducted on a ResNet-18 architecture with convolutional backbones; evaluating REDA on larger vision transformers (ViTs) and language models remains an important future direction. Attention-based models may experience representation mismatch differently than CNNs, requiring tailored activation calibration approaches. Second, REDA requires a tiny calibration dataset of N samples per task. While we demonstrated outstanding sample efficiency (even at $N = 16$), a zero-shot joint calibration remains an open research question. Devel-

oping generative or synthetic data techniques to perform joint calibration without real user data would further expand the utility of REDA in privacy-sensitive domains.

Furthermore, we discuss the societal impact and computational efficiency of our method. Model merging in general dramatically reduces the carbon footprint associated with deep learning by avoiding full-scale multi-task joint pre-training. By providing a highly effective calibration and alignment tool, REDA lowers the barrier to deploying multi-task systems. In terms of compute efficiency, both N-TAAC and Head adaptation are extremely lightweight, requiring only a single pass for calibration and 15 epochs of head-only optimization, which runs in a matter of seconds even on standard commodity hardware. This highlights the practical viability of our framework for edge deployment.

8. Conclusion and Future Work

In this work, we deconstructed representation and decision-boundary alignment for model merging. Through a massive, multi-seed empirical sweep across 97 configurations, we proved that backbone calibration (N-TAAC) and classification head adaptation are highly synergistic. Our dual-phase REDA framework sequentializes both to resolve variance collapse and align decision boundaries, unlocking superior multi-task performance and exceptional sample efficiency. Future work will investigate zero-shot calibration methods, apply REDA to large language models (LLMs) and diffusion experts, and explore other calibration subsets.

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A. Detailed Experimental Settings and Hyperparameters

To ensure complete transparency and facilitate exact replication of our empirical results, we document our detailed training and alignment configurations below.

A.1. Expert Fine-Tuning Settings

We utilize a standard ResNet-18 model family from the torchvision library initialized with ImageNet-pretrained weights. For task-specific expert training, we independent fine-tune the shared backbone on 3,000 randomly selected training images per task. Grayscale images in MNIST and Fashion-MNIST are duplicated across 3 channels to establish RGB compatibility and resized to 32×32 pixels. The exact optimization parameters for expert fine-tuning are as follows:

- Optimizer: AdamW (Loshchilov & Hutter, 2017)
- Learning Rate: 1×10^{-3}
- Weight Decay: 1×10^{-2}
- Learning Rate Schedule: Cosine Annealing with $T_{\max} = 5$ epochs
- Batch Size: 64
- Training Epochs: 5

During training, we only optimize the shared backbone and the task-specific classification head, keeping the other tasks’ heads frozen to prevent any gradient leakage. The final expert models reach individual test accuracies of 97.07% (MNIST), 85.06% (Fashion-MNIST), and 65.79% (CIFAR-10).

A.2. Head Adaptation Settings

For Phase 2 of our REDA framework (both Supervised Head SFT and Unsupervised Head TTA), we freeze the calibrated backbone and optimize only the parameters of the task-specific classification heads. The optimization parameters are:

- Optimizer: AdamW (Loshchilov & Hutter, 2017)
- Learning Rate: 1×10^{-3}
- Weight Decay: 0.0 (classification head only)
- Batch Size: 32
- Training Epochs: 15

Under Head TTA, we optimize the student head parameters by minimizing the KL-divergence distillation loss against the frozen specialized expert teacher. The KL-divergence loss is calculated using PyTorch’s native ‘KLDivLoss’ in ‘batchmean’ reduction mode:

$$\mathcal{L}_{\text{TTA}} = \text{KLDivLoss}(\log \text{Softmax}(z_{\text{student}}), \text{Softmax}(z_{\text{teacher}})) \quad (12)$$

B. Additional Empirical Material and Individual Task Performance

To provide further empirical transparency, we break down individual task accuracies across all three independent random seeds (seeds 42, 100, and 2026) for key configurations.

Table 3 details how REDA consistently outperforms the baselines across every seed and every individual task. Notably, on the highly complex CIFAR-10 task, simple Weight Averaging collapses to 32.04%, and Head TTA

alone achieves 43.31%. Sequential REDA (N-TAAC + TTA) recovers performance to 45.29%, demonstrating that representation calibration stabilizes deep features for complex natural image categories.

Similarly, on MNIST, uncalibrated Weight Averaging drops to 57.80%. Head TTA alone recovers to 78.39%, whereas REDA (N-TAAC + TTA) achieves a stellar 84.59%. This consistent performance boost across tasks of varying visual complexities highlights the generalizability of our proposed synergistic alignment.

Table 3. Task-by-task accuracy breakdown (%) across individual random seeds ($N = 128$).

Seed	Merge Method	Rep. Cal.	Head Align.	MNIST	F-MNIST	CIFAR-10
Seed 42	WA	NONE	NONE	57.80	28.39	32.04
	WA	NONE	TTA	78.12	69.59	43.06
	WA	NTAAC	NONE	72.53	62.96	40.51
	WA	NTAAC	TTA	85.25	72.42	43.91
Seed 100	WA	NONE	NONE	57.80	28.39	32.04
	WA	NONE	TTA	76.62	70.10	43.14
	WA	NTAAC	NONE	72.93	61.02	40.03
	WA	NTAAC	TTA	83.41	73.08	46.38
Seed 2026	WA	NONE	NONE	57.80	28.39	32.04
	WA	NONE	TTA	80.43	70.43	43.73
	WA	NTAAC	NONE	72.76	61.87	40.24
	WA	NTAAC	TTA	85.11	72.00	45.58

C. Hyperparameter Sensitivity Sweep of Head Adaptation

Under the empirical research philosophy of The Empiricist, we conduct a comprehensive hyperparameter sensitivity sweep to map the convergence properties, stability, and data efficiency of our dual-phase alignment framework. Specifically, we evaluate the interaction of learning rates $\eta \in \{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}\}$ and adaptation epochs $E \in \{10, 15, 20\}$ under Weight Averaging with native task-agnostic activation calibration (N-TAAC) on $N = 128$ samples (seed 42).

Tables 4 and 5 summarize the performance of REDA-SFT and REDA-TTA across these configurations.

C.1. Supervised Head Fine-Tuning (REDA-SFT) Analysis

As shown in Table 4, REDA-SFT is highly sensitive to both the learning rate and training epochs. At a low learning rate of $\eta = 1 \times 10^{-4}$, the classification head converges slowly, starting at 63.79% (10 epochs) and climbing steadily to 65.15% at 20 epochs. This suggests that while slow learning is stable, it requires more epochs to achieve near-optimal results.

Crucially, REDA-SFT reaches its peak average accuracy of 66.57% at a learning rate of $\eta = 5 \times 10^{-4}$ with 10 epochs, and stabilizes around 66.54% at 15 epochs. However, when the learning rate is set to our default $\eta = 1 \times 10^{-3}$ or higher ($\eta = 2 \times 10^{-3}$), we observe a slight performance degradation as the epoch count increases. For instance, at $\eta = 1 \times 10^{-3}$, average accuracy drops from 66.12% (10 epochs) to 65.70% (20 epochs). At $\eta = 2 \times 10^{-3}$, performance collapses further to 65.16% (20 epochs).

This behavior is a classic signature of overfitting on small data. Because our calibration dataset consists of only $N = 128$ samples per task, aggressive learning rates and excessive training epochs cause the classification head to memorize the tiny calibration set, sacrificing generalization on the full test sets. This highlights the importance of choosing a moderate learning rate (e.g., 5×10^{-4}) and a conservative epoch budget for supervised head fine-tuning under low-data regimes.

C.2. Unsupervised Test-Time Head Adaptation (REDA-TTA) Analysis

In contrast to the supervised SFT baseline, Table 5 demonstrates that unsupervised test-time adaptation (REDA-TTA) is remarkably robust, stable, and less prone to overfitting across the entire hyperparameter grid.

At a low learning rate of $\eta = 1 \times 10^{-4}$, REDA-TTA climbs steadily from 64.13% (10 epochs) to 65.43% (20

epochs). At $\eta = 5 \times 10^{-4}$, it reaches 67.13% at 20 epochs. At $\eta = 1 \times 10^{-3}$, it scales exceptionally well, achieving 67.19% at 15 epochs and peaking at 67.24% at 20 epochs. Even at a larger learning rate of $\eta = 2 \times 10^{-3}$, REDA-TTA remains highly effective, peaking at 67.16% (15 epochs) and only dropping slightly to 67.04% at 20 epochs.

This outstanding stability occurs because REDA-TTA optimizes the classification head using knowledge distillation from the specialized expert teachers rather than direct supervised label fitting. Distillation operates on soft-label distributions, providing a rich, high-entropy regularization signal that inherently prevents overfitting and stabilizes optimization, even on tiny data budgets. These empirical results firmly establish REDA-TTA as a highly robust and practical solution for post-merging multi-task alignment.

Table 4. REDA-SFT hyperparameter sensitivity sweep accuracies (%) under Weight Averaging ($N = 128$, seed 42).

Learning Rate (η)	Epochs (E)	MNIST	F-MNIST	CIFAR-10	Mean Accuracy (%)
1×10^{-4}	10	80.74	67.64	42.98	63.79
1×10^{-4}	15	81.48	68.58	43.59	64.55
1×10^{-4}	20	82.18	69.28	43.99	65.15
5×10^{-4}	10	83.82	71.28	44.60	66.57
5×10^{-4}	15	84.40	71.13	44.09	66.54
5×10^{-4}	20	84.69	71.09	43.59	66.46
1×10^{-3}	10	84.40	70.84	43.12	66.12
1×10^{-3}	15	84.69	70.92	42.67	66.09
1×10^{-3}	20	84.67	70.63	41.80	65.70
2×10^{-3}	10	84.27	69.97	41.64	65.29
2×10^{-3}	15	84.23	70.43	40.92	65.19
2×10^{-3}	20	84.53	70.05	40.90	65.16

Table 5. REDA-TTA hyperparameter sensitivity sweep accuracies (%) under Weight Averaging ($N = 128$, seed 42).

Learning Rate (η)	Epochs (E)	MNIST	F-MNIST	CIFAR-10	Mean Accuracy (%)
1×10^{-4}	10	80.75	68.30	43.33	64.13
1×10^{-4}	15	81.55	69.32	43.80	64.89
1×10^{-4}	20	82.24	70.19	43.85	65.43
5×10^{-4}	10	84.11	71.79	44.08	66.66
5×10^{-4}	15	84.63	72.27	44.17	67.02
5×10^{-4}	20	84.88	72.39	44.13	67.13
1×10^{-3}	10	84.81	71.88	44.06	66.92
1×10^{-3}	15	85.25	72.42	43.91	67.19
1×10^{-3}	20	85.46	72.48	43.79	67.24
2×10^{-3}	10	85.20	71.74	43.59	66.84
2×10^{-3}	15	85.33	72.64	43.52	67.16
2×10^{-3}	20	85.61	72.28	43.23	67.04

D. Empirical Verification of Variance Collapse

To empirically validate our mathematical deconstruction in Section 3.1 and demonstrate the severe activation-level distortion under model merging, we compute the standard deviation of activations at every BatchNorm layer of ResNet-18. We compare three models on the MNIST calibration set ($N = 128$, seed 42): the specialized MNIST expert, the uncalibrated Weight Averaging (WA) model, and our N-TAAC calibrated model.

Table 6 presents the layer-wise standard deviations. The empirical results demonstrate a profound “variance collapse” in the deeper layers of the uncalibrated merged model. While the standard deviation in the first layer of the merged model remains at 96.24% of the original expert (0.2644 vs. 0.2747), it decays progressively through sequential layers, reaching only 39.17% in Layer 19 (0.0376 vs. 0.0961) and 40.11% in Layer 20 (0.5205 vs. 1.2978).

This is an absolute textbook demonstration of compounding representation distortion across non-linear layers.

Crucially, our N-TAAC calibrated model successfully restores these statistics. In Layer 19, the standard deviation is boosted from 0.0376 to 0.0834 (86.78% of the expert). In Layer 20, the standard deviation is restored from 0.5205 to 1.1560 (89.07% of the expert). This empirical evidence confirms that N-TAAC acts as an essential representation stabilizer, preparing a high-quality backbone representation that enables stable and data-efficient classification head adaptation.

Table 6. Layer-wise activation standard deviations on the MNIST calibration set ($N = 128$, seed 42) for the specialized MNIST Expert, uncalibrated Weight Averaging (WA), and N-TAAC calibrated models.

Layer	MNIST Expert	WA Merged (Uncal)	REDA (N-TAAC Cal)	Ratio (WA / Expert)
Layer 1	0.2747	0.2644	0.2668	96.24%
Layer 2	0.1689	0.1688	0.1681	99.95%
Layer 3	0.4542	0.4409	0.4431	97.07%
Layer 4	0.1625	0.1554	0.1617	95.58%
Layer 5	0.5653	0.5324	0.5465	94.17%
Layer 6	0.1708	0.1648	0.1770	96.49%
Layer 7	0.2626	0.2548	0.2642	97.01%
Layer 8	0.2395	0.2170	0.2277	90.62%
Layer 9	0.1248	0.1251	0.1320	100.23%
Layer 10	0.2989	0.2899	0.3058	97.00%
Layer 11	0.1569	0.1443	0.1529	91.99%
Layer 12	0.2079	0.1820	0.1911	87.53%
Layer 13	0.1274	0.1256	0.1295	98.63%
Layer 14	0.0986	0.0723	0.0874	73.39%
Layer 15	0.2347	0.1791	0.2058	76.29%
Layer 16	0.0894	0.0517	0.0754	57.82%
Layer 17	0.2387	0.1260	0.1957	52.79%
Layer 18	0.2759	0.2166	0.2384	78.53%
Layer 19	0.0961	0.0376	0.0834	39.17%
Layer 20	1.2978	0.5205	1.1560	40.11%

E. Detailed Task Arithmetic Scaling Sensitivity Results

To complement the analysis in Section 5.4, we document the precise mean accuracies and standard deviations across all three independent random seeds ($N = 128$) under varying Task Arithmetic scaling coefficients in Table 7.

Table 7. Task Arithmetic scaling coefficient λ sensitivity sweep results. We report the mean multi-task average accuracy (%) and standard deviation in parentheses across three independent random seeds ($N = 128$).

λ	Uncalibrated TA (%)	N-TAAC alone (%)	Head TTA alone (%)	REDA (Ours) (%)
0.0	11.01 (± 0.00)	14.12 (± 0.14)	42.26 (± 0.56)	46.52 (± 1.25)
0.1	12.21 (± 0.00)	31.30 (± 0.34)	48.64 (± 0.70)	54.84 (± 0.86)
0.2	18.68 (± 0.00)	48.23 (± 0.36)	56.22 (± 0.77)	61.58 (± 0.78)
0.3	33.93 (± 0.00)	56.49 (± 0.38)	62.26 (± 0.62)	66.40 (± 0.56)
0.4	47.19 (± 0.00)	61.05 (± 0.38)	66.63 (± 0.25)	69.24 (± 0.48)
0.5	52.90 (± 0.00)	63.63 (± 0.39)	68.75 (± 0.18)	71.13 (± 0.37)
0.6	53.27 (± 0.00)	64.95 (± 0.45)	69.29 (± 0.20)	72.14 (± 0.28)
0.7	50.74 (± 0.00)	65.72 (± 0.41)	68.65 (± 0.36)	72.51 (± 0.25)
0.8	45.55 (± 0.00)	65.88 (± 0.39)	67.61 (± 0.21)	72.61 (± 0.27)
0.9	39.03 (± 0.00)	65.35 (± 0.39)	66.30 (± 0.16)	72.34 (± 0.22)
1.0	32.93 (± 0.00)	64.47 (± 0.34)	64.93 (± 0.23)	71.94 (± 0.19)
1.1	28.03 (± 0.00)	63.43 (± 0.27)	63.53 (± 0.14)	71.38 (± 0.22)
1.2	24.28 (± 0.00)	62.34 (± 0.33)	61.96 (± 0.14)	70.82 (± 0.29)
1.3	22.71 (± 0.00)	61.33 (± 0.38)	59.95 (± 0.19)	70.04 (± 0.30)
1.4	23.17 (± 0.00)	60.16 (± 0.36)	57.42 (± 0.13)	69.24 (± 0.29)
1.5	23.60 (± 0.00)	59.07 (± 0.36)	54.38 (± 0.43)	68.54 (± 0.22)